



Name: Tuba Jan

Student ID: CSC-25F-602

Subject: Object Oriented Programming

Teacher Name: Sir Sunil Kumar

Department: BSCS IIC (E)

Q1: Analog & Digital Training System (M21-7000)

The **M21-7000** is an all-in-one educational tool designed to help learners and experiment with electronics. Its main purpose is to provide a platform where you can build, test, and troubleshoot both **analog** (like amplifiers) and **digital** (like logic gates) circuits without needing separate expensive laboratory equipment.

2 Basic Safety Rules:

- 1. **Power Off Before Wiring:** Always turn off the main power switch before connecting or changing any wires on the breadboard to avoid short circuits.
- 2. **Check Polarity:** Before turning the system on, double-check that components like capacitors and LEDs are connected in the correct direction (+ and -) to prevent them from burning out.

Q2: Logic Gates and Truth Tables

1. NOT Gate

A **NOT Gate** is an inverter. It simply flips the input signal to its opposite.

- **What it does:** If the input is **High (1)**, the output is **Low (0)**. If the input is **Low (0)**, the output is **High (1)**.

Truth Table:

Input (A)	Output (Y)
0	1
1	0

2. AND Gate

An **AND Gate** acts like a security lock that requires two keys to open.

- **Condition for Output 1:** An AND Gate gives an output of 1 only when **all of its inputs are 1**. If even one input is 0, the output will be 0.

Truth Table (2-Input):

Input A	Input B	Output (Y)
0	0	0
0	1	0
1	0	0
1	1	1